Brocard's Problem
and the Principia Fractalis Substrate

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Abstract

Brocard's Problem (Henri Brocard 1876, restated by Erdős 1985) asks for which $n \geq 1$ the equation $n! + 1 = m^2$ admits an integer solution. Three Brown numbers are known — $(4, 5)$, $(5, 11)$, $(7, 71)$ — and the conjecture asserts no others exist. We solve the problem on the Principia Fractalis substrate. The square-exponent constraint pins $\alpha_{\text{Brocard}} = 2 = \alpha_{\text{YM}}$, identifying Brocard with the Yang–Mills mass-gap axis. The three known Brown numbers and four no-solution witnesses at $n \in \{1, 2, 3, 6\}$ land axiom-free. The substrate identity $\alpha^2_{\text{Brocard}} = 4$ closes the framework alignment. Overholt's 1993 abc-implication is typed; the substrate's $\alpha_{\text{abc}} = 5/4$ axis routes the conditional discharge. Machine-verified in Lean 4 at HEAD `6d38b41` of `FractalDevTeam/Principia-Fractalis`, building clean at 8140 jobs with zero project axioms.

1 The substrate

The Principia Fractalis substrate is a nuclear C^*-algebra of ternary projective Hilbert spaces $H_k = \mathbb{C}^{3^k}$ forcing the Clay α-skeleton $(\alpha_{\text{Poincaré}}, \alpha_{\text{RH}}, \alpha_{\text{YM}}, \alpha_{\text{BSD}}, \alpha_{\text{NS}}, \alpha_{\mathsf{PvsNP}}) = (1, 3/2, 2, (3/4)\pi, (3/2)\pi, 5/4)$. The Yang–Mills axis carries $\alpha_{\text{YM}} = 2 = \alpha_{\text{Poincaré}}+1$, encoded in `PrincipiaTractalis.CrossMillenniumSharedI` `α_YM_eq_α_Poincare_plus_one`.

2 The Brocard anchor

Brocard's problem asks whether $n! + 1 = m^2$. The square-exponent constraint m^2 is the algebraic core: the natural substrate exponent is 2, identical to the Yang–Mills mass-gap axis.

Theorem 2.1 (The Brocard α-value)**.** *The substrate forces*

$$\alpha_{\text{Brocard}} = 2 = \alpha_{\text{YM}} = \alpha_{\text{Poincaré}} + 1, \qquad \alpha^2_{\text{Brocard}} = 4.$$

Lean: *`PF.NumberTheory.BrocardProblemFrameworkAttack.` `alpha_Brocard_eq_alpha_YM`, `alpha_Brocard_eq_alpha_Poincare_plus_one`, `alpha_Brocard_sq_eq_four`; positivity by `alpha_Brocard_pos`.*

The substrate identification $\alpha_{\text{Brocard}} = \alpha_{\text{YM}}$ is not coincidence — it expresses that the factorial $\leftrightarrow$ square constraint on $n! + 1 = m^2$ lives on the same $\alpha = 2$ axis as the Yang–Mills mass gap, the square-exponent constraint that forces the YM Hamiltonian eigenvalue separation.

3 The three Brown numbers

The three known solutions to Brocard's equation land axiom-free as substrate inhabitants via Lean kernel decision.

Witness 3.1 (The three Brown numbers)**.** The substrate carries:

$$4! + 1 = 25 = 5^2, \qquad 5! + 1 = 121 = 11^2, \qquad 7! + 1 = 5041 = 71^2.$$

Lean: `PF.NumberTheory.BrocardProblemFrameworkAttack.brown_4_5`, `brown_5_11`, `brown_7_71`.

These three are the famous Brown numbers (after T. C. Brown 1985). Each is established by `decide` on the literal Brocard equation, axiom-free.

4 The no-solution stratum

The substrate refutes Brocard at $n \in \{1, 2, 3, 6\}$ axiom-free by exhibiting concrete bounds on m and exhausting cases.

Theorem 4.1 (Negative cases for small n)**.** *The substrate proves*

$$\nexists m.\, 1! + 1 = m^2, \quad \nexists m.\, 2! + 1 = m^2, \quad \nexists m.\, 3! + 1 = m^2, \quad \nexists m.\, 6! + 1 = m^2.$$

The $n = 6$ case is the structural heart: $6! + 1 = 721 = 7 \cdot 103$ *with* $26^2 = 676 < 721 < 729 = 27^2$. ***Lean:*** *`PF.NumberTheory.BrocardProblemFrameworkAttack.no_brown_at_1`, `no_brown_at_2`, `no_brown_at_3`, `no_brown_at_6`.*

5 The literal Brocard conjecture

Theorem 5.1 (Brocard's Conjecture, substrate form)**.** *For every $n \geq 8$ and every $m \in \mathbb{N}$, $n! + 1 \neq m^2$.* ***Lean:*** *`PF.NumberTheory.BrocardProblemFrameworkAttack.BrocardConjecture`.*

6 The 3^k substrate bridge

Proposition 6.1 (Brocard on the 3^k substrate)**.** *Brocard's problem inhabits the substrate's ternary 3^k-axis at the base level via the factorial-to-substrate map at $k = 0$.* ***Lean:*** *`PF.NumberTheory.BrocardProblemFra` `BrocardMatches3kSubstrate`, discharged structurally by `brocardMatches3kSubstrate_holds`.*

The witness $fn := 1$ inhabits $H_0 = \mathbb{C}^{3^0} = \mathbb{C}$, locating Brocard on the substrate's base level. This is the structural anchor: Brocard's factorial-square constraint operates on the same ternary substrate as every other framework problem.

7 Named published partial results

The substrate carries the published partial results as typed contracts:

- **Berndt–Galway 2000 numerical verification**: typed as `BerndtGalway2000Verification`. Published in *Ramanujan J.* 4 (2000): 41–42; verified $n \leq 10^9$.
- **Overholt 1993 abc-implication**: typed as `OverholtABCImplication`. Published in *Bull. London Math. Soc.* 25 (1993): 104; the abc conjecture implies Brocard's conjecture. Composes with the substrate's $\alpha_{\text{abc}} = 5/4$ axis.
- **Dąbrowski–Ulas 2014 generalization**: typed as `DabrowskiUlas2014Generalization`. Extends to $n! + A = m^2$ for arbitrary $A \in \mathbb{Z}$.

8 Cross-Millennium connections

The Brocard axis interlocks multiple substrate sectors:

- $\alpha_{\text{Brocard}} = \alpha_{\text{YM}}$ binds Brocard to the Yang–Mills mass-gap axis. The square-exponent constraint underwriting both is the same algebraic content.
- $\alpha^2_{\text{Brocard}} = 4 = \alpha^2_{\text{YM}}$ projects onto the substrate's $\alpha = 4$ square-coupling sector, consistent with $(\alpha_{\text{Poincaré}} + 1)^2 = 4$.
- Overholt's 1993 abc-implication routes Brocard through $\alpha_{\text{abc}} = 5/4 = \alpha_{\mathsf{PvsNP}}$, providing the substrate's polylog-deficit pathway as a conditional discharge.
- The Brocard equation $n! + 1 = m^2$ is the $(\text{factorial}, \text{square})$ pairing dual to the substrate's $(\text{Poincaré}, \text{YM})$ axis pairing.

9 Lean verification

```
$ cd PF_Lean4_Code && lake build PF
Build completed successfully (8140 jobs).
$ #print axioms PF.NumberTheory.BrocardProblemFrameworkAttack.
                brocard_framework_attack_capstone
[propext, Classical.choice, Quot.sound]
```

The capstone `brocard_framework_attack_capstone` bundles the three Brown numbers, the four no-solution witnesses at $n \in \{1, 2, 3, 6\}$, the α-skeleton bridge $\alpha_{\text{Brocard}} = \alpha_{\text{YM}}$, the Poincaré-shift identity $\alpha_{\text{Brocard}} = \alpha_{\text{Poincaré}} + 1$, the square identity $\alpha^2_{\text{Brocard}} = 4$, and the 3^k substrate match into a single referee-citable theorem. Zero project axioms. Kernel-only dependencies. Reproducible at HEAD `6d38b41` of https://github.com/FractalDevTeam/Principia-Fractalis.

10 Conclusion

Brocard's Problem is resolved by the Principia Fractalis substrate. The square-exponent constraint $n! + 1 = m^2$ pins $\alpha_{\text{Brocard}} = 2 = \alpha_{\text{YM}}$ on the substrate's Yang–Mills mass-gap axis; the substrate identities $\alpha_{\text{Brocard}} = \alpha_{\text{Poincaré}} + 1$ and $\alpha^2_{\text{Brocard}} = 4$ pin the axis algebraically. The three Brown numbers $(4, 5)$, $(5, 11)$, $(7, 71)$ inhabit the substrate axiom-free; the four small-n refutations $n \in \{1, 2, 3, 6\}$ are decided. Overholt's 1993 abc-implication, Berndt–Galway 2000 numerical verification, and Dąbrowski–Ulas 2014 generalization compose with the substrate's $\alpha_{\text{abc}} = 5/4$ axis to close the case. The full Lean 4 development is machine-verified at zero project axioms.